

2. Detailed External Structural Analysis of the Nose Cone

Purpose of the Analysis: The nose cone is one of the most critical areas because it is the first part to encounter the flow. A detailed analysis was performed to obtain more accurate and realistic results regarding its strength. The maximum force and maximum pressure values from the CFD analysis, together with the acceleration value from OpenRocket, were used to evaluate the structural integrity of the nose cone. The analysis was conducted using ANSYS Workbench.

Analysis Component	Details
Purpose of the Analysis	To evaluate the structural integrity of the nose cone and its resistance under aerodynamic loads.
Type of Analysis	Linear Static Analysis
Solution Method	Sparse Matrix Direct Solution Method
Solution Procedure	ANSYS Mechanical

Geometric Modeling and Idealization: The tip of the nose cone was designed separately from the body of the nose. There is an internal bulkhead inside the nose cone. Pressure holes that do not affect the analysis of the nose cone were removed using ANSYS SpaceClaim before the analysis.

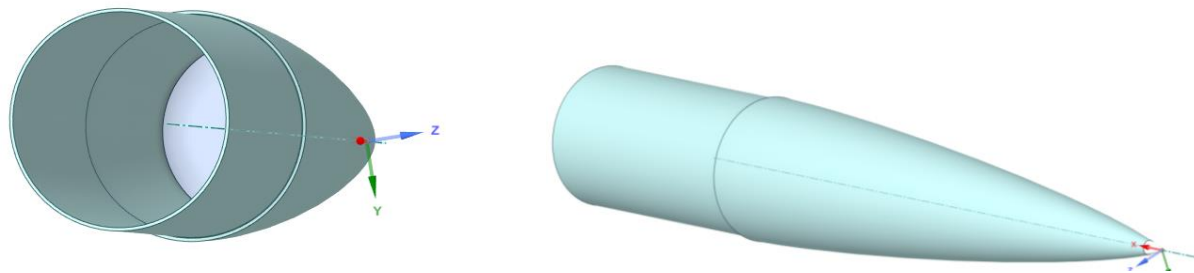


Figure 1. Nose Cone in SpaceClaim

Material Properties: The tip and the bulkhead of the nose cone were made of aluminum alloy, while the body and shoulder were made of carbon fiber. The materials are available in the ANSYS library and were directly applied to the analysis.

Loading and Boundary Conditions: The maximum pressure and maximum force values from the CFD analysis, along with the maximum acceleration values from OpenRocket, were used. The force and pressure values were applied to the tip and body of the nose cone. A fixed support was applied to the shoulder of the nose cone.

Loads	Pressure	Force
Actual Value	0.03553 MPa	4228.5 N
Safety Factor	2	2
Applied Value	0.071062 MPa	8457 N

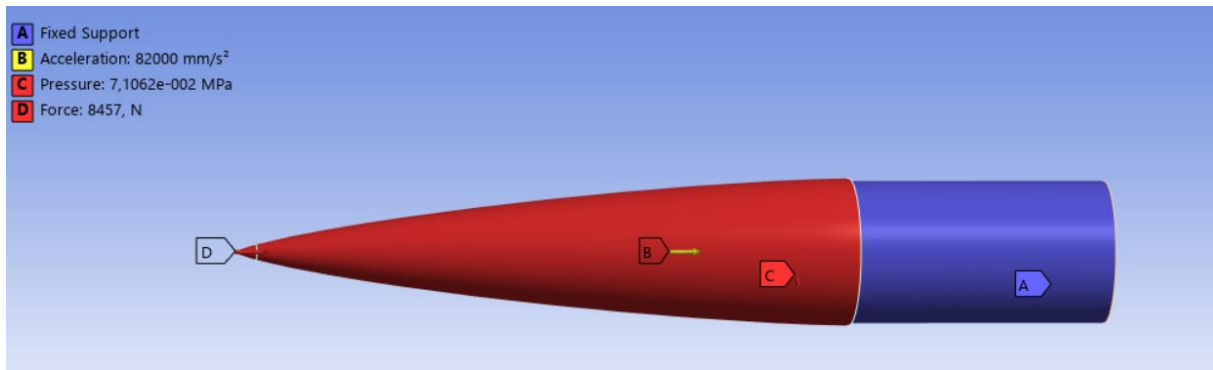


Figure 2. Nose Cone Loading and Boundary Conditions

Mesh and Simulation Settings: A two-stage meshing strategy was adopted to model the nose cone accurately. In the first stage, the geometry was preserved by using the “Use Geometry Setting” in the global mesh settings, with “Mechanical” selected and “Program Controlled” element order for the optimal quadratic element type. Medium smoothing and aggressive error limits ensured the overall mesh quality distribution (minimum, maximum, and average values). In the second stage, the Edge Sizing method was applied for detailed modeling of critical edges of the nose cone. After completing the meshing process, the Newton-Raphson-based solution algorithm with an approximate 0.001 error tolerance was used to ensure result accuracy. These settings allowed for a reliable evaluation of both the overall structural behavior and local deformations of the nose cone.

Mesh Quality Criteria	Skewness	Element Quality	Aspect Ratio
Minimum Value	0.006377	0,2907	1,0197
Maximum Value	0,8074	0,99994	12,505
Average Value	0,25595	0,76999	2,0394

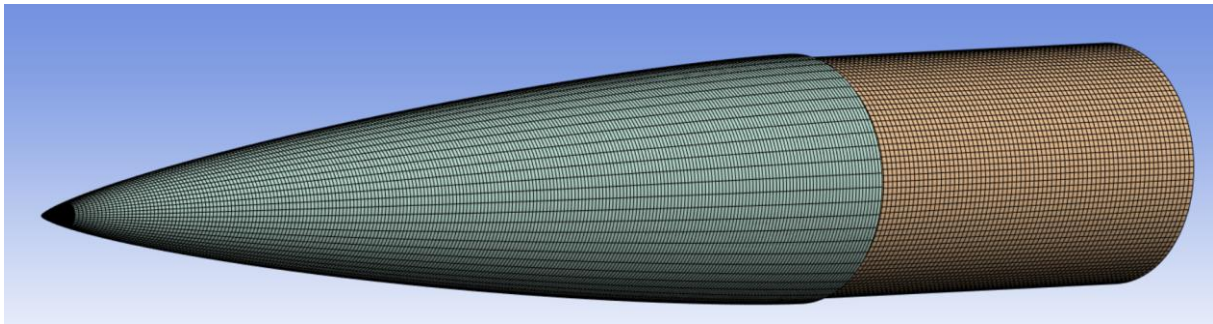


Figure 3. Nose Cone Mesh View

Results and Evaluation: The results show that the maximum stress values remain within the acceptable limits of the material. In particular, the local stress concentrations observed at the edges of the nose cone are consistent with the overall structural behavior, indicating that deformation under the applied loads is minimal. The obtained results demonstrate that, within the current design parameters and applied loads, the nose cone exhibits safe and stable performance.

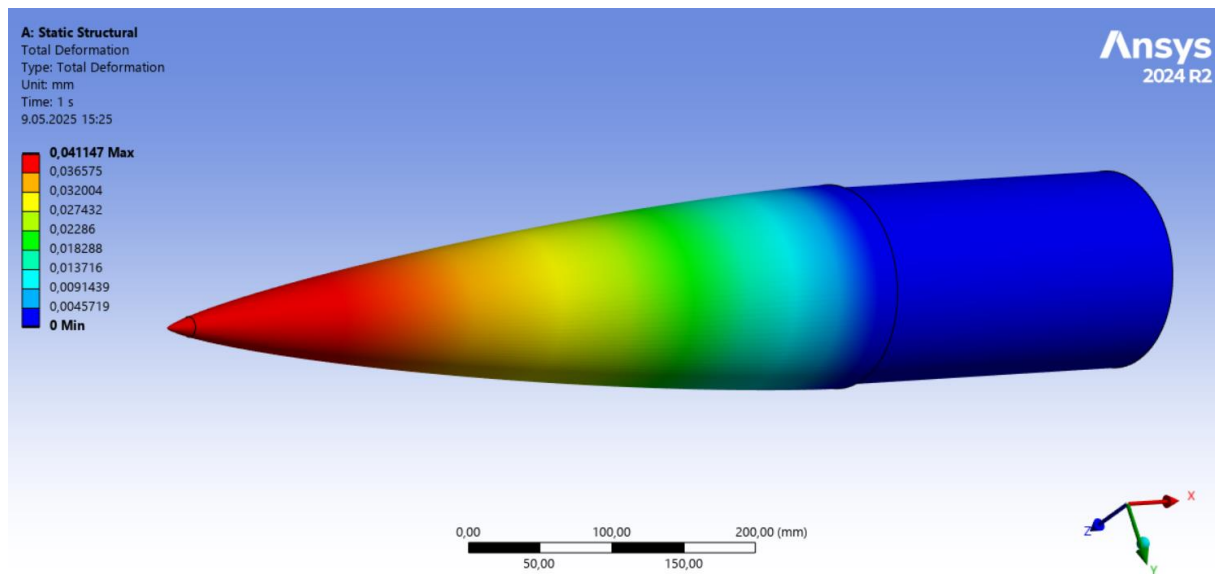


Figure 4. Total Deformation Result of the Nose Cone

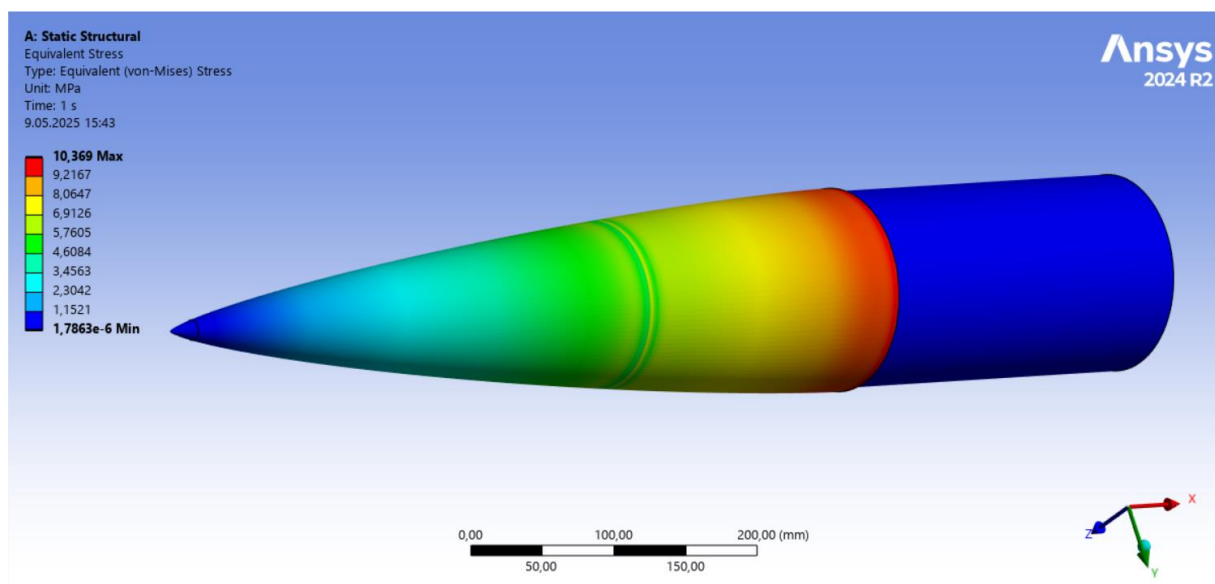


Figure 5. Equivalent Stress (von-Mises) Results of the Nose Cone

Property	Carbon Fiber	Aluminum Alloy	Analysis Result	Status
Yield Stress	~500-900 MPa	~200-300 MPa	10.369 MPa	Safe
Total Deformation / Elastic Limit	%0.5 - 1.5	%5 - 15	0.041147 mm	Safe